EXPERIMENT 2

Academic Year: 2025-26 **Year and Sem: BEVIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: To implement an item-based nearest neighbor recommendation system using Python

Theory: Collaborative Filtering (CF) is a popular technique in recommender systems that predicts user preferences based on past interactions. It assumes that users who agreed in the past will agree in the future. CF is broadly classified into:

- **User-based CF:** Finds similar users and recommends items they liked.
- **Item-based CF:** Finds similar items and recommends them to the user.

Item-Based CF focuses on the **relationship between items** rather than users. If a user liked item A, they are likely to enjoy items similar to A.

Key Concepts

a) User-Item Interaction Matrix

This is a matrix where rows represent items and columns represent users. Each cell contains the rating given by a user to an item, or 0 if unrated.

b) Item Similarity

Item similarity is computed to find how closely two items are related. Common measures include:

1. **Cosine Similarity:** Measures the cosine of the angle between two item vectors.
2. **Pearson Correlation:** Measures linear correlation between item ratings.
3. **Adjusted Cosine Similarity:** Accounts for user rating bias by subtracting user mean ratings.

Cosine Similarity Formula:

$$\text{sim}(i,j) = \frac{\sum_{u \in U} r_{u,i} \cdot r_{u,j}}{\sqrt{\sum_{u \in U} r_{u,i}^2} \cdot \sqrt{\sum_{u \in U} r_{u,j}^2}}$$

Where:

- $R_{u,i}$ = rating of user u on item i
- U = set of all users

Making Recommendations

Once item similarities are computed:

1. Identify items that the user has rated.
2. For each rated item, find similar items that the user has not rated.
3. Compute a weighted score for each candidate item based on similarity and the user's rating.
4. Recommend top N items with the highest scores.

Weighted Prediction Formula:

$$\hat{r}_{u,i} = \frac{\sum_{j \in S(i)} \text{sim}(i,j) \cdot r_{u,j}}{\sum_{j \in S(i)} |\text{sim}(i,j)|}$$

Where S(i) is the set of items similar to item i.

Applications

1. E-commerce: Suggesting products similar to purchased items (Amazon).
2. Streaming platforms: Recommending movies or songs similar to liked ones (Netflix, Spotify).
3. Online learning: Suggesting courses similar to completed courses.

Conclusion

In this experiment, an item-based nearest neighbor recommendation system was implemented using user–movie rating data. The dataset was successfully preprocessed by merging ratings with movie information for effective analysis. Item similarities were computed to generate personalized movie recommendations for users.